Towhid Ahmed

Generative AI Engineer · Open to remote — US/EU time zones
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Professional Summary

Generative AI Engineer with a solid **ML/DL foundation**. Expert in architecting **agentic workflows** and **RAG pipelines** using **LangChain** and **FastAPI**. Focused on mitigating hallucinations, **optimizing retrieval**, and **deploying scalable**, **low-latency** AI solutions via **Docker** and **cloud** platforms. **Proven ability to ship production-grade systems independently** — purpose-built for high-velocity, remote-first teams.

Technical Skills

Programming Languages & Data: Python (Expert), Java (Data Structures & Algorithms), PostgreSQL

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, EDA, Data Preprocessing, Feature Engineering, Random Forest, **TensorFlow**, **Scikit-learn**, **CNN**, RNN/LSTM, Model Evaluation

NLP & Generative AI: LLMs, **Prompt Engineering**, **RAG Pipelines**, **Vector Databases** (FAISS, ChromaDB, Weaviate), Sentence Transformers, **Cross-Encoder Re-ranking**, **LangChain**, LLaMA, HuggingFace, BERT, Transformers, **Multi-Agent Systems**, **AI Agents**.

MLOps & Deployment: MLflow, DVC, DagsHub, **Docker**, **FastAPI**, Streamlit, **REST APIs**, **CI/CD**, **Render**, Railway.

Software Engineering: 400+ DSA problems solved, Object-Oriented Programming (OOP), System Design

Developer Tools: Pandas, NumPy, Next.js, GitHub, Claude, GitHub Copilot, Cursor AI, Groq API, Hugging Face Hub

Technical Experience

Advanced RAG-Based ML Research Assistant — Weaviate | Sentence Transformers | OpenAI API | Groq | Streamlit | Render (**GitHub**)

- Built an **end-to-end RAG pipeline** for semantic search and Q&A over arXiv ML research papers using **Weaviate** vector **database** and OpenAI/Groq LLMs.
- Implemented multiple **document chunking strategies** (fixed-size, sentence-based,semantic) with sentence-transformer embeddings for high-quality vector indexing.
- Improved retrieval accuracy using **cross-encoder re-ranking** and evaluated system performance with **5 key metrics** — Precision, Recall, MRR, NDCG, and F1 Score.
- **Deployed** the Streamlit app on **Render** with a production-ready cloud setup.

Multi-Agent AI System — Python | LangChain | Groq | Tavily API | BeautifulSoup | Streamlit (**Demo | GitHub**)

- Built a **multi-agent AI pipeline** that autonomously searches the web, scrapes content, generates structured research reports, and **self-critiques output quality** using LangChain and Groq LLM.
- Designed **modular agents** — Search Agent (Tavily API), Reader Agent (BeautifulSoup scraping), Writer Chain (structured report generation), and **Critic Chain** (quality scoring and feedback).
- Developed an end-to-end CLI pipeline and an interactive Streamlit UI for exploring the full research workflow on any user-defined topic.

Medical Assistant — LLM-Powered Chatbot — Python | FastAPI | LangChain | ChromaDB | Groq | HuggingFace | Tavily API (**Demo | GitHub**)

- Architected a **dual-retrieval RAG pipeline** combining local **ChromaDB** (MMR) and Tavily web search scoped to authoritative medical sources (NIH, WHO, Mayo Clinic), with **automated emergency symptom escalation**.
- Engineered a **stateful LangChain inference pipeline** with **sliding-window conversation memory (k=8)**, HuggingFace embeddings, and per-response source attribution grounded in retrieved context.
- Exposed a **multi-modal document ingestion system** via **REST API**, UI upload, and folder watch — chunking and indexing PDFs/TXTs into a persistent vector store at runtime.

AI IDE — Multi-Agent Coding Assistant | Python | FastAPI | Groq | FAISS | React | Render (Demo | GitHub)

- Built a Cursor-like AI coding assistant with a **multi-agent backend** (Planner, Coder, Debugger, Researcher) using Groq's **llama-3.3-70b**, real-time **SSE token streaming**, and a Monaco Editor frontend – deployed on Render.
- Engineered a RAG pipeline (**FAISS + SentenceTransformers**) for context-aware codebase memory and an **11-endpoint REST API** covering the full coding workflow — generation, debugging, refactoring, and planning.
- Integrated Tavily web search into the Research Agent for real-time, AI-synthesized answers grounded in live results.

Education and Certificates

EDUCATION: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology, Bangladesh

CERTIFICATIONS: ● **Mathematics for ML & Generative AI: Linear Algebra, Probability, Statistics, Calculus — Udemy**

- **DSA**, **Machine Learning & Deep Learning** – CodeBasics
- **MLOps** – Udemy ● **Generative AI** – Udemy